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Convert the insights into simple, jargon-free, actionable recommendations the business can use immediately.

This is your Python notebook

Start solving by adding a new cell below — write and run code, or jot notes in text cells as you go.

Analyzing Basic Metrics

[ ]
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

[ ]
```
#Load the dataset
df=pd.read_csv('aerofit_treadmill.csv')
df.head()
#Shape
print("Shape of the dataset",df.shape)
```

[ ]
```
#Info / dtypes / missing values
df.info()
df.isnull().sum()
```

[ ]
```
#unique values
df.nunique()
```

[ ]
```
#Convert discrete/categorical columns to 'category' dtype
cat_cols = ['Product', 'Gender', 'MaritalStatus', 'Usage', 'Fitness']
for col in cat_cols:
    df[col] = df[col].astype('category')

df.dtypes
```

[ ]
```
#Statistical summary
#Numerical columns
df.describe()
```

[ ]
```
#categorical column
df.describe(include='category')
```

[ ]
```
df.nunique()
```

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```
    print("\nProportion:")
    print(df[col].value_counts(normalize=True).round(3))
    print()
```

Insights to note:

*The "typical" AeroFit customer profile emerging: male, partnered, uses the treadmill 3x/week, moderate self-rated fitness (3/5), and most likely to buy the entry-level KP281.
*Usage and Fitness are both concentrated around the middle values (3), with a right-skew toward higher usage/fitness — consistent with the smaller premium-buyer segment (KP781) seen earlier.
*Age, Income, Miles, and Education have far more unique values, confirming they behave as continuous/near-continuous variables rather than categories — good justification for keeping them numeric rather than converting to category.

[ ]
```
#3.Visual Analysis
#Correlation heatmap
df_enc = df.copy()
df_enc['Product_enc'] =
df_enc['Product'].map({'KP281':1,'KP481':2,'KP781':3})
df_enc['Gender_enc'] = df_enc['Gender'].map({'Female':0,'Male':1})
df_enc['MaritalStatus_enc'] =
df_enc['MaritalStatus'].map({'Single':0,'Partnered':1})

num_cols = ['Age','Education','Usage','Fitness','Income','Miles',
            'Gender_enc','MaritalStatus_enc','Product_enc']

plt.figure(figsize=(9,7))
sns.heatmap(df_enc[num_cols].corr(), annot=True, fmt='.2f',
cmap='coolwarm', square=True)
plt.title('Correlation Heatmap')
plt.show()
```

Insight:Product_enc correlates most with Income (0.62), Fitness (0.59), Miles (0.57), Usage (0.54), and Gender_enc (0.47 — more males buy pricier models).
MaritalStatus_enc is ~0 → no relationship with product choice, confirming earlier finding.

[ ]
```
#pairplot
sns.pairplot(df, vars=['Age','Income','Usage','Fitness','Miles'],
hue='Product',
             palette='viridis', diag_kind='hist')
plt.show()
```

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```
        sns.histplot(df[col], kde=True, ax=ax, bins=15)
        ax.axvline(df[col].mean(), color='red', linestyle='--', label='mean')
        ax.axvline(df[col].median(), color='black', label='median')
        ax.set_title(f'Distribution of {col}')
        ax.legend()
    plt.tight_layout(); plt.show()
```

```
[ ] #boxplot
    fig, axes = plt.subplots(1, 4, figsize=(18,4.5))
    for ax, col in zip(axes, ['Age','Income','Usage','Miles']):
        sns.boxplot(data=df, x='Product', y=col, hue='Product', legend=False,
    ax=ax)
        ax.set_title(f'{col} by Product')
    plt.tight_layout(); plt.show()
```

```
[ ] #boxplot-income vs other categorical features
    fig, axes = plt.subplots(1, 3, figsize=(15,4.5))
    sns.boxplot(data=df, x='Gender', y='Income', hue='Gender', legend=False,
    ax=axes[0])
    sns.boxplot(data=df, x='MaritalStatus', y='Income', hue='MaritalStatus',
    legend=False, ax=axes[1])
    sns.boxplot(data=df, x='Fitness', y='Income', hue='Fitness', legend=False,
    ax=axes[2])
    plt.tight_layout(); plt.show()
```

```
[ ] #Effect of Marital Status & Age on Product
    df['AgeGroup'] = pd.cut(df['Age'], bins=[17,24,29,34,50], labels=['18-
    24','25-29','30-34','35-50'])

    fig, axes = plt.subplots(2, 2, figsize=(14,10))
    sns.countplot(data=df, x='Product', hue='MaritalStatus', ax=axes[0,0])
    pd.crosstab(df['Product'], df['MaritalStatus'],
    normalize='index').plot(kind='bar', stacked=True, ax=axes[0,1])
    sns.countplot(data=df, x='AgeGroup', hue='Product', ax=axes[1,0])
    pd.crosstab(df['AgeGroup'], df['Product'],
    normalize='index').plot(kind='bar', stacked=True, ax=axes[1,1])
    plt.tight_layout(); plt.show()
```

Conclusion: Age group shows mild fluctuation (e.g., 25–29 buys KP781 more, 30–34 leans KP481) but no strong monotonic trend — consistent with the near-zero raw correlation (0.03) found earlier. Marital status and age are weak-to-no predictors of product choice; Income, Fitness, Usage, and Miles remain the real drivers.

4.Missing Value & Outlier Detection

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```
fig, axes = plt.subplots(1, 4, figsize=(16,4.5))
for ax, col in zip(axes, cont_cols):
    sns.boxplot(y=df[col], ax=ax, color='skyblue')
    ax.set_title(f'Boxplot: {col}')
plt.tight_layout()
plt.show()#Detect Outliers
```

[ ]
```
# describe() + mean vs median gap
desc = df[cont_cols].describe().T
desc['mean_minus_median'] = desc['mean'] - desc['50%']
print(desc[['mean','50%','std','min','max','mean_minus_median']])
```

[ ]
```
# IQR method - count actual outliers per column
for col in cont_cols:
    Q1, Q3 = df[col].quantile(0.25), df[col].quantile(0.75)
    IQR = Q3 - Q1
    lower, upper = Q1 - 1.5*IQR, Q3 + 1.5*IQR
    n_out = ((df[col] < lower) | (df[col] > upper)).sum()
    print(f"{col}: bounds=({lower:.1f}, {upper:.1f}), outliers={n_out}")
```

[ ]
```
#Missing values
df.isnull().sum()
```

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Insights:- **No missing values** across all 180 records — dataset is complete, no cleaning needed.
19 Income outliers and 13 Miles outliers (via IQR) are the main concern; Age and Education have far fewer (4–5).
**Mean > Median for Age, Income, and Miles confirms right-skew — a small group of high-value customers pulls the average up.
These outliers aren't errors — they're mostly KP781 buyers** (high income, high mileage), representing AeroFit's genuine premium-athlete segment.
Don't delete them; keep for customer profiling, but use median (not mean) or winsorize when reporting/modeling to avoid overstating the "typical" customer.

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5.Probability & Customer Profiling

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```
P(Product | Gender)
ct_col = pd.crosstab(df['Gender'], df['Product'], normalize='columns') #
P(Gender | Product)
```

P(KP781 | Male) = 33/104 = 0.317 (31.7%)

[ ]
```
#Marginal Probability of Product
df['Product'].value_counts(normalize=True)
```

[ ]
```
#Normalized Crosstabs — Marginal & Conditional Probabilities
for feat in ['MaritalStatus','Gender','Fitness','Usage']:
    print(pd.crosstab(df[feat], df['Product'], normalize='index').round(3))
```

These conditional probabilities are far more predictive than the flat marginal — e.g., P(KP781) jumps from 22.2% overall to 93.5% once you condition on Fitness=5.

[ ]
```
#Customer Profiling by Product
df['Usage'] = df['Usage'].astype(int)
df['Fitness'] = df['Fitness'].astype(int)

df.groupby('Product')
[['Age','Income','Usage','Fitness','Miles','Education']].mean().round(1)
```

Insights & Conclusion
Gender and fitness/usage are strong predictors, marital status is not — P(KP781|Male)=31.7% vs P(KP781|Female)=9.2%, while marital status shows almost identical splits across all three products.
Conditional probability massively outperforms the marginal for prediction — the overall P(KP781)=22.2%, but conditioning on Fitness=5 or Usage≥6 pushes it to 93–100%, meaning fitness/usage level should drive real-time recommendations far more than a generic base rate.
KP281 and KP481 share one customer profile; KP781 is a genuinely distinct segment — the entry and mid-tier models attract the same type of buyer (differentiated mainly by price, not persona), while KP781 buyers are a clearly separate high-income, high-performance cluster.
Actionable takeaway: AeroFit should build its recommendation logic around Fitness rating, weekly Usage, and Income (not Age or Marital Status) — and should specifically target more female high-fitness customers for KP781, since they're currently underrepresented (9.2%) relative to their male counterparts (31.7%) despite comparable fitness potential.

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mid-20s to early 30s.

Income: Ranges $29,562–$104,581, typical customer earns $44K–$59K (median $50.6K). A smaller group earns well above $80K.

Education: Ranges 12–21 years, typical customer has 14–16 years (roughly college-level), median 16.

Usage (times/week): Ranges 2–7, typical customer uses it 2–4 times/week (median 3).

Fitness (self-rated 1–5): Ranges 1–5, most common rating is 3 (moderate), with a second smaller peak at 5 (very fit).

Miles (weekly): Ranges 21–360, typical customer runs 66–115 miles/week (median 94). A small group runs well over 200.

Distribution Shapes & Relationships

Age, Income, and Miles are all right-skewed (mean > median) — most customers cluster at moderate values, with a smaller group of older/wealthier/high-mileage customers pulling the tail to the right. Miles is the most skewed of the three.

Education is close to symmetric, slightly left-leaning (most people cluster at 14–16 years, few below).

Heatmap: Income, Fitness, Usage, and Miles all move together — customers with higher income tend to also report higher fitness and log more miles. Age is essentially unrelated to everything else (correlation ≈ 0).

Pairplot: KP781 buyers form a visibly separate cluster at high Income/Fitness/Miles, while KP281 and KP481 buyers overlap almost completely — meaning the split isn't 3 distinct customer types, it's really "everyday users" vs. "serious athletes."

Univariate Plot Comments

1. Product Countplot

sns.countplot(x='Product')

KP281 has the tallest bar (80 customers), KP481 is second (60), and KP781 is shortest (40). This shows AeroFit's sales volume is heaviest at the entry-level price point and drops off as price increases.

2. Age Histogram

sns.histplot(df['Age'], kde=True)

The bars are tallest between 22–28 and taper off gradually toward 50, with almost nothing above 45. This right-skewed shape (mean 28.8 > median 26) means most customers are in their mid-20s, with a small tail of older buyers.

3. Income Histogram

sns.histplot(df['Income'], kde=True)

Most bars sit between $40,000–$55,000, with a long thin tail stretching out to $104,581. The shape is clearly right-skewed, telling us the "typical" AeroFit customer earns around $50K, but a small group of much wealthier customers exists.

4. Miles Histogram

sns.histplot(df['Miles'], kde=True)

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treadmill.

5. Gender Countplot
sns.countplot(x='Gender')
The Male bar (104) is noticeably taller than Female (76). Overall, the customer base skews male by about 16 percentage points.

6. Fitness Countplot
sns.countplot(x='Fitness')
Rating "3" has by far the tallest bar (97 customers), with a second, smaller bar at "5" (31 customers). This bimodal pattern hints at two customer types: an average-fitness majority and a smaller high-fitness minority.

7. Usage Countplot
sns.countplot(x='Usage')
Bar heights peak at "3" times/week (69 customers) and "4" times/week (52), then drop sharply for 5, 6, and 7. Most customers plan moderate usage; very few plan heavy daily-level usage.

Bivariate Plot Comments

8. Income vs Product Boxplot
sns.boxplot(x='Product', y='Income')
The KP281 and KP481 boxes sit low and overlap almost entirely (~$46K–$49K median). The KP781 box sits much higher (~$76K median) with almost no overlap with the other two — income clearly separates KP781 buyers from the rest.

9. Fitness vs Product Boxplot
sns.boxplot(x='Product', y='Fitness')
KP281 and KP481 boxes cluster tightly around fitness level 3. The KP781 box sits almost entirely at level 4–5 with very little overlap — self-rated fitness is one of the strongest visual separators between product tiers.

10. Usage vs Product Boxplot
sns.boxplot(x='Product', y='Usage')
Similar pattern to Fitness: KP281/KP481 boxes sit around 3 uses/week, while the KP781 box sits around 5, with its lower whisker still above the median of the other two — heavier planned usage strongly predicts a KP781 purchase.

11. Age vs Product Boxplot
sns.boxplot(x='Product', y='Age')
All three boxes overlap heavily and sit at roughly the same height (median 26–27 across all products). Unlike Income/Fitness/Usage, Age shows no visual separation between product buyers.

12. Gender vs Product Countplot (stacked/grouped)
sns.countplot(x='Product', hue='Gender')
For KP281 the Male and Female bars are equal height (40 each). For KP481 they're close (31 vs 29). For KP781, the Male bar is nearly 5x taller than the Female bar (33 vs 7) — gender imbalance is only visually obvious at the premium tier.

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(roughly 60/40 each time) — visually, marital status doesn't shift the product mix at all.

14. Correlation Heatmap
sns.heatmap(corr, annot=True)
The darkest (strongest) cells connecting to Product are Income (0.62), Fitness (0.59), and Miles (0.57) — all shown in warm colors. The Age-Product cell is pale/near-white (0.03), visually confirming Age has almost no linear relationship with product choice.

15. Pairplot (Income, Fitness, Miles, Usage colored by Product)
sns.pairplot(df, vars=[...], hue='Product')
In the Income-vs-Miles and Fitness-vs-Miles scatter panels, the KP781-colored points form a visually separate cluster in the upper-right (high income, high miles, high fitness), while KP281 and KP481 points overlap heavily in the lower-left — this confirms KP781 buyers form a distinct group rather than just a random subset of all buyers.

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